Lecture 3: Numeral Systems

CMPS 221 – Computer Organization and Design

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Numeral Systems

- A **numeral system** is a notation for representing numbers

- It consists of:
  - A set of *symbols* each representing a value
  - A set of *rules* for combining symbols to represent other values

Roman Numeral System

Symbols

I	one
V	five
X	ten
L	fifty
C	hundred
D	five hundred
M	one thousand

MMCXLIV

$\downarrow$

$$M + M + C - X + L - I + V$$

(two thousand one hundred forty four)

Arabic Numeral System

Symbols

0	zero
1	one
2	two
3	three
4	four
5	five
6	six
7	seven
8	eight
9	nine

$$2735$$

$$\downarrow$$

$$2\times10^3 + 7\times10^2 + 3\times10^1 + 5\times10^0$$

Ten is an important number here!

- We have *ten* symbols whose values corresponding to the first *ten* natural numbers
- Symbols are combined to represent other values by multiplying the value of each symbol by *ten* to the power of its position and adding them

Another name for this system is the **decimal** system

Positional Numeral Systems

- In general, a **positional numeral system** with a **base** *B* is a numeral system where:
  - *B* symbols are used whose values correspond to the first *B* natural numbers
  - Symbols are combined to represent other values by multiplying the value of each symbol by *B* to the power of its position and adding them

$$s_n s_{n-1} \ldots s_1 s_0$$

$$\downarrow$$

$$s_n \times B^n + s_{n-1} \times B^{n-1} + \ldots + s_1 \times B^1 + s_0 \times B^0$$

- The decimal system is a positional numeral system with base ten

A Numeral System for Computers

- Computers are comprised of circuits that store charge and process electric signals in one of *two* states: high voltage or low voltage
- For this reason, a numeral system for computers ought to have only *two* symbols (0 and 1)
- Numbers in computers are represented using a positional numeral system with base two
  - Called the **binary** numeral system

Binary Numeral System Examples

1101_2
$= 1 \times 2^3 + 1 \times 2^2 + 0 \times 2^1 + 1 \times 2^0$
$= 8 + 4 + 1$
$= 13_{10}$

110101_2
$= 1 \times 2^5 + 1 \times 2^4 + 0 \times 2^3 + 1 \times 2^2 + 0 \times 2^1 + 1 \times 2^0$
$= 32 + 16 + 4 + 1$
$= 53_{10}$

Decimal	**Binary**
0_{10}	0_2
1_{10}	1_2
2_{10}	10_2
3_{10}	11_2
4_{10}	100_2
5_{10}	101_2
6_{10}	110_2
7_{10}	111_2
8_{10}	1000_2
9_{10}	1001_2
10_{10}	1010_2
11_{10}	1011_2
12_{10}	1100_2
13_{10}	1101_2
14_{10}	1110_2
15_{10}	1111_2

Conversion from Decimal to Binary

- Conversion from decimal to binary can be done by successively dividing the number by two and extracting the remainder

- Example: converting 94_{10} to binary

94 / 2 = 47	remainder 0	(least significant digit)
47 / 2 = 23	remainder 1	
23 / 2 = 11	remainder 1	
11 / 2 = 5	remainder 1	
5 / 2 = 2	remainder 1	
2 / 2 = 1	remainder 0	
1 / 2 = 0	remainder 1	(most significant digit)

Answer: 1011110_2

Other Numeral Systems

- Since binary can be verbose, computer scientists sometimes use other systems in writing for brevity
  - **Octal**: positional system with base 8
  - **Hexadecimal**: positional system with base 16

- Conversion between these systems is straightforward because 8 and 16 are powers of 2

Octal Numeral System Examples

4307_8
$= 4\times8^3 + 3\times8^2 + 0\times8^1 + 7\times8^0$
$= 2247_{10}$

1563_8
$= 1\times8^3 + 5\times8^2 + 6\times8^1 + 3\times8^0$
$= 883_{10}$

Decimal	Binary	Octal
0_{10}	0_2	0_8
1_{10}	1_2	1_8
2_{10}	10_2	2_8
3_{10}	11_2	3_8
4_{10}	100_2	4_8
5_{10}	101_2	5_8
6_{10}	110_2	6_8
7_{10}	111_2	7_8
8_{10}	1000_2	10_8
9_{10}	1001_2	11_8
10_{10}	1010_2	12_8
11_{10}	1011_2	13_8
12_{10}	1100_2	14_8
13_{10}	1101_2	15_8
14_{10}	1110_2	16_8
15_{10}	1111_2	17_8

Octal-Binary Conversion

- Every three binary digits (bits) correspond to one octal digit
- Examples:
  - Convert 10110101110001100_2 to octal

    010 110 101 110 001 100
    2 6 5 6 1 4   Answer: 265614_8

  - Convert 3176_8 into binary

    3 1 7 6
    011 001 111 110   Answer: 11001111110_2

Hexadecimal Numeral System Examples

$BC12_{16}$
$= B \times 16^3 + C \times 16^2 + 1 \times 16^1 + 2 \times 16^0$
$= 11 \times 16^3 + 12 \times 16^2 + 1 \times 16^1 + 2 \times 16^0$
$= 48146_{10}$

$CAFE_{16}$
$= C \times 16^3 + A \times 16^2 + F \times 16^1 + E \times 16^0$
$= 12 \times 16^3 + 10 \times 16^2 + 15 \times 16^1 + 14 \times 16^0$
$= 51966_{10}$

Use symbols {A,B,C,D,E,F} to represent natural numbers beyond nine

Decimal	**Binary**	**Octal**	**Hex.**
0_{10}	0_2	0_8	0_{16}
1_{10}	1_2	1_8	1_{16}
2_{10}	10_2	2_8	2_{16}
3_{10}	11_2	3_8	3_{16}
4_{10}	100_2	4_8	4_{16}
5_{10}	101_2	5_8	5_{16}
6_{10}	110_2	6_8	6_{16}
7_{10}	111_2	7_8	7_{16}
8_{10}	1000_2	10_8	8_{16}
9_{10}	1001_2	11_8	9_{16}
10_{10}	1010_2	12_8	A_{16}
11_{10}	1011_2	13_8	B_{16}
12_{10}	1100_2	14_8	C_{16}
13_{10}	1101_2	15_8	D_{16}
14_{10}	1110_2	16_8	E_{16}
15_{10}	1111_2	17_8	F_{16}

Hexadecimal-Binary Conversion

- Every four bits correspond to one hexadecimal digit

- Examples:
  - Convert 10110101110001100_2 to hexadecimal

    0001 0110 1011 1000 1100
    1 6 B 8 C Answer: $16B8C_{16}$

  - Convert $374F_{16}$ into binary

    3 7 4 F
    0011 0111 0100 1111 Answer: 0011011101001111_2

Addition

Decimal

```
   19
+  62
----
```

Addition

Decimal

```
carry   1
        19
      + 62
      ----
         1
```

Addition

Decimal

```
carry    1
         19
       + 62
       ----
         81
```

Addition

```
            Decimal         Binary

carry         1
             19             10011
           + 62            111110
           ----           -------
             81
```

Addition

```
             Decimal        Binary

carry          1
               19            10011
             + 62           111110
             ----          -------
               81                1
```

Addition

```
         Decimal      Binary

carry        1             1
            19         10011
          + 62        111110
          ----       -------
            81            01
```

Addition

```
          Decimal        Binary

carry        1              11
            19           10011
        +   62          111110
        ------        --------
            81             001
```

Addition

```
        Decimal      Binary

carry     1            111
          19         10011
        + 62        111110
        ----       -------
          81          0001
```

Addition

```
         Decimal         Binary

carry    1      1111
         19      10011
       + 62     111110
       ----    -------
         81      10001
```

Addition

```
            Decimal        Binary

carry          1         11111
              19          010011
            + 62          111110
            ----         -------
              81          010001
```

Addition

```
         Decimal      Binary

carry       1         11111
           19       0010011
         + 62       0111110
         ----       -------
           81       1010001
```

Subtraction

Decimal

```
    5  1
-   2  2
--------
```

Subtraction

Decimal

```
borrow    4
          ~~5~~  ¹1
-         2      2
--------------------
```

Subtraction

Decimal

```
borrow    4
          ~~5~~  ¹1
-         2   2
---------------
              9
```

Subtraction

Decimal

```
borrow     4
           ~~5~~  ¹1
-          2      2
--------------
           2      9
```

Subtraction

	Decimal	Binary
borrow	*4*	
	~~5~~ ^{1}1	0 1 1 0 0 1 1
-	2 2	0 0 1 0 1 1 0
	--------	---------------------
	2 9	

Subtraction

	Decimal			Binary						
borrow	*4*									
	~~5~~	[1]1	0	1	1	0	0	1	1
-	2	2	0	0	1	0	1	1	0
	2	9							1

Subtraction

		Decimal			Binary						
borrow		*4*									
		~~5~~	1 1	0	1	1	0	0	1	1	
	-	2	2	0	0	1	0	1	1	0	
		-------		---------------							
		2	9						0	1	

Subtraction

		Decimal		Binary						
borrow		*4*				*0*	*1*			
		~~5~~	^{1}1	0	1	~~1~~	~~0~~	^{1}0	1	1
	-	2	2	0	0	1	0	1	1	0
	--------			--------------------						
		2	9						0	1

Subtraction

		Decimal					Binary			
borrow		*4*				*0*	*1*			
		~~5~~	^{1}1	0	1	~~1~~	~~0~~	^{1}0	1	1
	-	2	2	0	0	1	0	1	1	0
		2	9					1	0	1

Subtraction

	Decimal			Binary						
borrow	*4*				*0*	*1*				
	~~5~~	11	0	1	~~1~~	~~0~~	10	1	1	
-	2	2	0	0	1	0	1	1	0	
-------			-------------------							
	2	9				1	1	0	1	

Subtraction

Decimal

borrow	*4*	
	~~5~~	[1]1
-	2	2
	2	9

Binary

	0	*[1]0*	*1*			
0	~~1~~	~~1~~	~~0~~	[1]0	1	1
0	0	1	0	1	1	0
			1	1	0	1

Subtraction

	Decimal			Binary						
borrow	*4*				*0*	^{1}*0*	*1*			
	~~5~~	^{1}1	0		~~1~~	~~1~~	~~0~~	^{1}0	1	1
-	2	2	0		0	1	0	1	1	0
	--------		-------------------							
	2	9				1	1	1	0	1

Subtraction

	Decimal			Binary						
borrow	*4*				*0*	[1]*0*	*1*			
	~~5~~	[1]1	0	~~1~~	~~1~~	~~0~~	[1]0	1	1	
-	2	2	0	0	1	0	1	1	0	
	-------	-------	---	---	---	---	---	---	---	
	2	9		0	1	1	1	0	1	

Subtraction

		Decimal					Binary			
borrow		*4*			*0*	^{1}*0*	*1*			
		~~5~~	^{1}1	0	~~1~~	~~1~~	~~0~~	^{1}0	1	1
	-	2	2	0	0	1	0	1	1	0
		2	9	0	0	1	1	1	0	1

Borrowing is Complicated

- Subtraction of binary numbers via borrowing is complicated and inefficient to implement in hardware

- Instead, we use a convenient representation of negative binary numbers called *two's complement* that simplifies subtraction

Two's Complement

- The **two's complement** of an *N*-bit number is an *N*-bit number such that the sum of the numbers is 2^N
  - Example: The two's complement of 0010110 is 1101010

```
   0010110
 + 1101010
 ---------
  10000000 = 2^7
```

- Two's complement is convenient for representing negative numbers because:
  - The sum of a number and it's two's complement is zero if the $(N+1)^{th}$ digit is ignored
  - Leftmost digit is 0 if positive and 1 if negative

Finding Two's Complement

- Shortcut for finding two's complement:
  - Invert all the bits then add 1

$$-x = \bar{x} + 1$$

- Example:

Original number:	`0010110`
Invert all bits:	`1101001`
Add one:	`1101010`

Subtraction

	Decimal			Binary							
borrow		*4*									
		~~5~~	[1]1		0	1	1	0	0	1	1
	-	2	2	-	0	0	1	0	1	1	0
	--------			-----------------------							
		2	9								

Subtraction with Two's Complement

	Decimal	Binary
borrow	*4*	
	~~5~~ [1]1	0 1 1 0 0 1 1
	- 2 2	+ 1 1 0 1 0 1 0
	--------	--------------------
	2 9	

Subtraction with Two's Complement

	Decimal				Binary						
borrow	*4*										
	~~5~~	[1]1		0	1	1	0	0	1	1
−	2	2	+	1	1	0	1	0	1	0
	2	9								1

Subtraction with Two's Complement

	Decimal			Binary							
borrow	*4*								*1*		*carry*
	~~5~~	1 1		0	1	1	0	0	1	1	
-	2	2	+	1	1	0	1	0	1	0	
	2	9							0	1	

Subtraction with Two’s Complement

	Decimal			Binary								
borrow		*4*								*1*		*carry*
		~~5~~	1 1		0	1	1	0	0	1	1	
	−	2	2	+	1	1	0	1	0	1	0	
	--------			----------------------								
		2	9						1	0	1	

Subtraction with Two's Complement

	Decimal			**Binary**							
borrow	*4*								*1*		*carry*
	$\cancel{5}$	$^{1}1$		0	1	1	0	0	1	1	
−	2	2	+	1	1	0	1	0	1	0	
	2	9					1	1	0	1	

Subtraction with Two's Complement

	Decimal			Binary								
borrow		*4*							*1*		*carry*
		~~5~~	1 1		0	1	1	0	0	1	1
	−	2	2	+	1	1	0	1	0	1	0
	--------			----------------------							
		2	9				1	1	1	0	1

Subtraction with Two's Complement

```
                Decimal                      Binary

borrow          4              1                       1  carry
              ~~5~~ ¹1         0   1   1   0   0   1   1
         -      2   2      +   1   1   0   1   0   1   0
         --------------    ---------------------------
                2   9              0   1   1   1   0   1
```

Subtraction with Two’s Complement

		Decimal			Binary							
borrow		*4*			*1*				*1*			*carry*
		~~5~~	11		0	1	1	0	0	1	1	
	−	2	2	+	1	1	0	1	0	1	0	
	--------			----------------------								
		2	9	~~1~~	0	0	1	1	1	0	1	

Multiplication

Decimal

```
  23
× 13
----
```

Multiplication

Decimal

```
   23
×  13
----
   69

----
```

Multiplication

Decimal

```
   23
×  13
----
   69
  23
----
```

Multiplication

Decimal

```
   23
×  13
----
   69
  23
----
  299
```

Multiplication

Decimal

```
  23
× 13
----
  69
 23
----
 299
```

Binary

```
 10111
  1101
------
```

Multiplication

Decimal

```
   23
×  13
----
   69
  23
----
  299
```

Binary

```
    10111
     1101
  -------
    10111



---------
```

Multiplication

Decimal

```
  23
× 13
----
  69
 23
----
 299
```

Binary

```
    10111
     1101
   ------
    10111
   00000

---------
```

Multiplication

Decimal

```
   23
×  13
 ----
   69
  23
 ----
  299
```

Binary

```
   10111
    1101
  ------
   10111
  00000
 10111

---------
```

Multiplication

Decimal

```
  23
× 13
----
  69
 23
----
 299
```

Binary

```
     10111
      1101
    ------
     10111
    00000
   10111
  10111
 ---------
```

Multiplication

Decimal	Binary

```
   23              10111
×  13               1101
 ----             ------
   69              10111
  23              00000
 ----            10111
  299           10111
               ---------
               100101011
```

Division

Decimal

$$4\overline{)54}$$

Division

Decimal

$$4 \overline{\smash{)}\,54} \quad \text{quotient so far: } 1$$

Division

Decimal

$$\begin{array}{r|l} & \;1 \\ \hline 4 & 54 \\ & 4 \end{array}$$

Division

Decimal

$$\begin{array}{r|l} & \ \ 1 \\ \cline{2-2} 4 & \ \ 5\,4 \\ & \ \ \underline{4} \\ & \ \ 1 \end{array}$$

Division

Decimal

$$
\begin{array}{r|l}
 & \ 1 \\
\cline{2-2}
4 & 54 \\
 & \underline{4} \\
 & 14
\end{array}
$$

Division

Decimal

$$
\begin{array}{r|l}
 & \;\;13 \\ \hline
4 & \;\;54 \\
 & \;\;\underline{4} \\
 & \;\;14
\end{array}
$$

Division

Decimal

$$
\begin{array}{r|l}
 & 13 \\ \hline
4 & 54 \\
 & \underline{4} \\
 & 14 \\
 & 12
\end{array}
$$

Division

Decimal

$$
\begin{array}{rl}
 & 13 \\ \cline{2-2}
4 & \big) 54 \\
 & \underline{4} \\
 & 14 \\
 & \underline{12} \\
 & 2
\end{array}
$$

Division

Decimal

$$\begin{array}{rl} & 13 \\ 4 & \overline{)54} \\ & \underline{4} \\ & 14 \\ & \underline{12} \\ & 2 \end{array}$$

Binary

$$\begin{array}{rl} 100 & \overline{)110110} \end{array}$$

Division

Decimal	Binary

Decimal

```
     13
  ┌────
4 │ 54
    4
   ──
    14
    12
   ───
     2
```

Binary

```
          1
    ┌───────
100 │110110
```

Division

Decimal	Binary
$$\begin{array}{r} 13 \\ 4\overline{)54} \\ \underline{4} \\ 14 \\ \underline{12} \\ 2 \end{array}$$	$$\begin{array}{l} 1 \\ 100\,\overline{)110110} \\ 100 \end{array}$$

Division

Decimal

$$
\begin{array}{r|l}
 & \;\;13 \\ \hline
4 & \;\;54 \\
 & \;\;\underline{4} \\
 & \;\;14 \\
 & \;\;\underline{12} \\
 & \;\;2
\end{array}
$$

Binary

$$
\begin{array}{r|l}
 & 1 \\ \hline
100 & 110110 \\
 & \underline{100} \\
 & 10
\end{array}
$$

Division

Decimal

```
    13
  ____
4 | 54
    4
    --
    14
    12
    --
     2
```

Binary

```
          1
    _______
100 |110110
     100
     ---
      101
```

Division

Decimal

```
     13
   ____
4 | 54
    4
    _
    14
    12
    __
     2
```

Binary

```
          11
     _______
100 | 110110
      100
      ___
        101
```

Division

Decimal	Binary

$$
\begin{array}{r|l}
 & 13 \\ \hline
4 & 54 \\
 & \underline{4} \\
 & 14 \\
 & \underline{12} \\
 & 2
\end{array}
\qquad\qquad
\begin{array}{r|l}
 & 11 \\ \hline
100 & 110110 \\
 & \underline{100} \\
 & 101 \\
 & 100
\end{array}
$$

Division

Decimal

```
     13
   ----
 4 | 54
     4
    --
     14
     12
     --
      2
```

Binary

```
          11
     -------
 100 |110110
      100
      ---
       101
       100
       ---
         1
```

Division

Decimal

```
    13
  ----
4 | 54
    4
    --
    14
    12
    --
     2
```

Binary

```
          11
     -------
100 | 110110
      100
      ---
       101
       100
       ---
         11
```

Division

Decimal

```
     13
   ____
4 | 54
    4
   --
    14
    12
   ---
     2
```

Binary

```
         110
     _______
100 | 110110
      100
      ---
       101
       100
       ---
         11
```

Division

Decimal	Binary

<pre>   13
4 ⟌54
   4
   14
   12
    2</pre>
	<pre> 110
	100 ⟌110110
	    100
	     101
	     100
	       110</pre>

Division

Decimal	Binary

Decimal

```
    13
  ----
4 | 54
    4
   --
    14
    12
   ---
     2
```

Binary

```
           1101
      ---------
100 | 110110
      100
      ---
       101
       100
       ---
         110
```

Division

Decimal

```
    13
  ----
4 | 54
    4
    --
    14
    12
    --
     2
```

Binary

```
        1101
    -------
100 | 110110
      100
      ---
       101
       100
       ---
         110
         100
```

Division

Decimal

```
     13
   ____
4 | 54
    4
    --
    14
    12
    --
     2
```

Binary

```
          1101
     _______
100 |110110
     100
     ---
      101
      100
      ---
        110
        100
        ---
         10
```

Textbook Sections

- Some of the content in these slides corresponds to:
  - Textbook:
    - *Computer Organization and Design, 5th Edition by David Patterson and John Hennessy, Morgan Kaufmann, 2014.*
  - Sections:
    - 2.4, 3.1-3.4 (excluding instructions and hardware which will be covered later)